Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/01/2017
PI Name:	Jaing, Crystal Ph.D.		
Project Title:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) Project		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Livermore	**State:**	CA
Zip Code:	94550-9698	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/01/2015	**End Date:**	04/30/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE change in grant number and end date per NSSC information and F. Hernandez/ARC; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019 (Ed., 1/18/2018) NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 8/22/17)		
Key Personnel Changes/Previous PI:	Feb 2016: No changes.		
COI Name (Institution):	Mehta, Satish Ph.D. (Wyle Laboratories, Inc.) Pierson, Duane Ph.D. (NASA Johnson Space Center) Smith, David (NASA Ames Research Center) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC18K0113 ; NNX15AJ29G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with various ISS modules using traditional culture-based methods and state-of-the art molecular techniques. The objectives of the ISS-MOP project are to measure presence of viral and select bacterial and fungal pathogens and correlate their presence on crew. The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. The hypothesis of the ISS-MOP project is that certain short-living viruses associated with inert environmental spaces in the closed habitat of the ISS can have an adverse influence on crew health. To test this hypothesis the team will catalog viruses associated with surfaces and air of the ISS modules using the all-encompassing Lawrence Livermore Microbial Detection Array (LLMDA), which contains probes to detect not only viruses but also clinically important human bacterial and fungal pathogens. In addition, skin and saliva samples from crewmembers will be requested to test the hypothesis. There are three aims proposed in this ISS-MOP project. While the cultivation-based assay (Aim 1) provides a valid estimation of microbial presence, it offers only a limited assessment of the phylogenetic/pathogenic diversity and physiological breadth of the microbial population present. Therefore, the team will use the novel and comprehensive LLMDA microarray, which is specifically designed to elucidate the pathogenic viral and microbial diversity profiles of complex samples to at least the species level (Aim 2). In addition, we will characterize the virulence and antibiotic resistance mechanisms associated with these pathogens from the ISS by developing a new Virulence and Antibiotic Resistance gene array and analyze the samples on this array. The final deliverable of the proposed project to NASA will be the ISS-MOP dataset, a compilation of all of the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat (Aim 3). The objectives of the ISS-MOP project directly respond to the National Research Council Committee for the Decadal Survey on Biological and Physical Sciences in Space recommendation P1 where it states, “to establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.” The ISS-MOP project will also provide answers to NASA Space Biology Guiding Questions: How do alterations in gravity affect in microbial cells (CMM1)?; Does the enhanced virulence or other genetic changes observed with microbial cultures flown in space occur in other species (CMM4)?; and What are the genetic changes underlying physiological responses to spaceflight demonstrated in biological systems (CMM5)?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with crewmembers and various ISS modules using both traditional culture-based methods and state-of-the art molecular techniques to a degree that has not previously been possible.
Task Progress:	During this performing period, we made progress on science definition and science readiness for our flight experiments. We evaluated two air samplers: NIOSH Personal Bioaerosol Cyclone Sampler (PBCS) and Sartorious air sampler (MD8). We selected MD8 due to higher biomass collected and higher efficiency. Additionally, we evaluated the DNA-free gelatinous filter as the filter to collect air samples and verified these filters as DNA free and capable of collecting viable bacteria and fungi. We tested human swab collection procedures and DNA yield and modified collection procures to maximize sample yield. We evaluated the Lawrence Livermore Microbial Detection Array and shotgun metagenomic sequencing tools using previously collected ISS filter and dust samples. This resulted in a manuscript recently submitted to Microbiome. We evaluated the detection of antibiotic resistance genes and virulence genes from DNA sequencing data from previously collected ISS samples. A manuscript is in preparation. We evaluated analysis of viral samples on the Lawrence Livermore Microbial Detection Array using several human clinical and spiked samples. Several herpesviruses were detected. We presented at Informed Consent Briefing meeting with the Inc 53/54 crew on Sept, 2016 and Inc 55/56 on Feb, 2017 to discuss the project goals and the samples to be collected preflight, inflight, and postflight. Our project enrolled three crew. The first crew’s preflight samples were collected on Nov 2016 and Jan 2017. We worked on IRB (Institutional Review Board) renewal and prepared a Crew Questionnaire and amended IRB on Jan 2017. Operationally, payload implementation was initiated at Ames Research Center (ARC), July 2016. The ERD and Mini-ED were signed in Nov 2016. Baseline Data Collection Plan for Inc 51/52 was signed on Dec 2016. Data Sharing Plan Document for Inc 51/52 was signed Jan 2017. Regarding project meetings, we have held monthly MT-2 meetings with ARC Project Scientist to update the team on schedule and upcoming milestones, review SVT, and readiness. MT-1 and -2 teams met with the GeneLab team to discuss project data submission in April 2016. ISS-MOP investigators meeting was conducted Feb 2017 at Jet Propulsion Laboratory (JPL) to discuss work progress and current and future research plans.
Bibliography Type:	Description: (Last Updated: 04/07/2023)
Abstracts for Journals and Proceedings	Checinska A, Frey KG, Redden CL, Thissen J, Allen JE, Jaing C, Wheeler K, Venkateswaran K. "Targeted Amplification of Antibiotic Resistant Genes Associated with the International Space Station Environment." Presented at the American Society of Microbiology 2016 meeting, Boston, MA, June 16-20, 2016. American Society of Microbiology 2016 meeting, Boston, MA, June 16-20, 2016. , Jun-2016
Abstracts for Journals and Proceedings	Jaing C, Smith D, Mehta S, Venkateswaran K. "Microbial Observatory of Pathogens in International Space Station: Bacteria, Fungi, and Viruses." Oral presentation at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016

Articles in Peer-reviewed Journals	Checinska Sielaff A, Singh NK, Allen JE, Thissen J, Jaing C, Venkateswaran K. "Draft genome sequences of biosafety level 2 opportunistic pathogens isolated from the environmental surfaces of the International Space Station." Genome Announc. 2016 Dec 29;4(6). http://dx.doi.org/10.1128/genomeA.01263-16 ; PubMed PMID: 28034853; PubMed Central PMCID: PMC5201052 , Dec-2016
Significant Media Coverage	Wampler S. (LLNL public affairs office) "LLNL biodetection system bound for space." Lawrence Livermore National Laboratory press release, April 28, 2016. https://www.llnl.gov/news/llnl-biodetection-system-bound-space , Apr-2016